FIITJEE

ALL INDIA INTEGRATED TEST SERIES

HALF COURSE TEST – I

JEE (Main)-2020

TEST DATE: 30-12-2018

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. C

Sol. According to Stefan-Boltzmann's Law, energy radiated per sec per unit area is:

$E = e\sigma(T^4 - T_0^4)$

Hence, unit of $\sigma = \dfrac{joule/\sec - m^2}{K^4} = \dfrac{watt}{m^2 \times K^4}$.

2. D

Sol. Here $(2\pi ct/\lambda)$ as well as $(2\pi x/\lambda)$ are dimensionless. So, unit of ct is same as that of λ.

Unit of x is same as that of λ.

Since, $\left[\dfrac{2\pi ct}{\lambda}\right] = \left[\dfrac{2\pi x}{\lambda}\right] = \left[M^0L^0T^0\right]$

Hence, $\dfrac{2\pi c}{\lambda} = \dfrac{2\pi x}{\lambda t}$

In the option $(d), \dfrac{x}{\lambda}$ is unitless. It is not the case with $\dfrac{c}{\lambda}$.

3. D

Sol. In $\cos\left(\dfrac{t}{p} - qx\right)$, the expression $\left(\dfrac{t}{p} - qx\right)$ is dimensionless. Hence, t/p is also dimensionless. This is possible when units of t are same as that of p.

4. D

Sol. $\left[\frac{mr^2}{6\pi\eta}\right]=\left[\frac{ML^2}{ML^{-1}T^{-1}}\right]=\left[L^3T\right]$

$[\eta]=\left[ML^{-1}T^{-1}\right]$

$\left[\left(\frac{6\pi mr\eta}{g^2}\right)^{1/2}\right]=\left[\left(\frac{MLML^{-1}T^{-1}}{L^2T^{-4}}\right)^{1/2}\right]=\left[ML^{-1}T^{3/2}\right]$

$\left[\frac{m}{6\pi\eta rv}\right]=\left[\frac{M}{ML^{-1}T^{-1}LLT^{-1}}\right]=\left[L^{-1}T^2\right]$

Thus, none of the given expression have the dimensions of time.

5. D

Sol. Maximum percentage error in the measurement of pressure,

$\frac{\Delta P}{P}\times 100=\frac{\Delta F}{F}\times 100+2\times\frac{\Delta r}{r}\times 100$

$=\frac{4}{100}\times 100+2\times\frac{2}{100}\times 100=8\%$

6. B

Sol. Here, maximum fractional error is:

$\frac{\Delta Q}{Q}=\pm\left(n\frac{\Delta X}{X}+m\frac{\Delta Y}{Y}\right)$

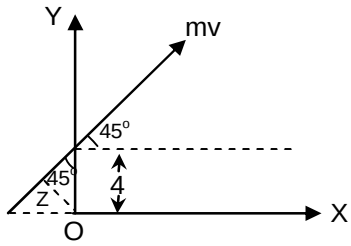
$\therefore$ Absolute error in Q, *i.e.*,

$\Delta Q=\pm\left(n\frac{\Delta X}{X}+m\frac{\Delta Y}{Y}\right)Q$

7. A

Sol. Momentum of the particle $=mass\times velocity$

$=(5)\times\left(3\sqrt{2}\right)=15\sqrt{2}$



The direction of momentum in the XOY plane is given by:

$y=x+4$

Slope of the line $=1=\tan\theta$

i.e., $\theta=45^o$

Intercept of this st. line = 4

Length of the perpendicular z from the origin on the straight line:

$=4\sin 45^o=\frac{4}{\sqrt{2}}=2\sqrt{2}$

Angular momentum = momentum $\times$ perpendicular length = $15\sqrt{2}\times 2\sqrt{2}=60\ unit$.

8. D

Sol. Momentum at the highest point

$=mV\cos 45^o=\frac{mV}{\sqrt{2}}$

Maximum height $= h$

$$h = \frac{V^2 \sin^2 45^o}{2g} = \frac{V^2}{4g}$$

Angular momentum $= L$

$$L = momentum \times height = \frac{mV}{\sqrt{2}} \times \frac{V^2}{4g} = \frac{mV^3}{4\sqrt{2}g}$$

But $\quad \frac{V^2}{4g} = h \quad or \quad V = \sqrt{4gh}$

$$L = \frac{m}{4\sqrt{2}} \frac{(4gh)^{3/2}}{g} = m\sqrt{2gh^3}\ .$$

9. B

Sol. $x = \frac{k}{b^2}(1 - e^{-bt})$

$$\frac{dx}{dt} = \frac{k}{b} e^{-bt},\ \frac{d^2x}{dt^2} = -ke^{-bt}$$

10. D

Sol. Here, $\frac{dv}{dt} = -kv^3$

or $\quad \frac{dv}{v^3} = -kdt \quad or \quad \int_{v_0}^{v} \frac{dv}{v^3} = \int_0^t -kdt$

$$\left[-\frac{1}{2v^2} \right]_{v_0}^{v} = -kt$$

or $\quad \frac{1}{2v_0^2} - \frac{1}{2v^2} = -kt$

$\therefore \quad \frac{1}{2v^2} = \frac{1 + 2v_0^2 kt}{2v_0^2} \quad or \quad v^2 = \frac{v_0^2}{1 + 2v_0^2 kt}$

$\therefore \quad v = \frac{v_0}{\sqrt{(2v_0^2 kt + 1)}}\ .$

11. D

Sol. $x = ae^{-\alpha t} + be^{\beta t}$

$$\frac{dx}{dt} = a\alpha e^{-\alpha t} + b\beta e^{\beta t}$$

$$v = -a\alpha e^{-\alpha t} + b\beta e^{\beta t}$$

12. B

Sol. $y = 12x - \frac{3}{4}x^2$

$$\frac{dy}{dt} = 12\frac{dx}{dt} - \frac{3}{2}x\frac{dx}{dt}$$

At $x = 0$: $\quad \frac{dy}{dt} = 12\frac{dx}{dt}$

If θ be the angle of projection, then

$$\frac{dy/dt}{dx/dt} = 12 = \tan\theta$$

Also, if u =initial velocity, then $u\cos\theta = 3$

Hence, $\tan\theta \times u\cos\theta = 36$ *or* $u\sin\theta = 36$

Range, $R = \frac{u^2 \sin 2\theta}{g} = \frac{2u^2 \sin\theta\cos\theta}{g}$

$$= \frac{2(u\sin\theta)(u\cos\theta)}{10} = \frac{2\times 36\times 3}{10} = 21.6\ m.$$

13. B

Sol. $h = (u\sin\theta)t - \frac{1}{2}gt^2$

$$d = (u\cos\theta)t$$

or $\quad t = \frac{d}{u\cos\theta}$

$\therefore \quad h = u\sin\theta.\frac{d}{u\cos\theta} - \frac{1}{2}g.\frac{d^2}{u^2\cos^2\theta}$

$\therefore \quad u = \frac{d}{\cos\theta}\sqrt{\frac{g}{2(d\tan\theta - h)}}$.

14. C

Sol. Given: $\frac{\sqrt{3}u}{2} = u\cos\theta =$ speed at maximum height

$\therefore \quad \cos\theta = \frac{\sqrt{3}}{2}$

or $\quad \theta = 30^o$

Given that; $pH_{\max.} = R$

We know, $H_{\max.} = \frac{R\tan\theta}{4}$

$\therefore \quad P = \frac{4}{\tan\theta} = \frac{4}{\tan 30^o} = 4\sqrt{3}$

15. C

Sol. In a conservative field, $F = -\frac{dU}{dr}$

$\therefore \quad F = -\frac{d}{dx}(ax^2 - bx) = b - 2ax$.

16. A

Sol. Net force on the rod = $F_1 - F_2 \quad (\because F_1 > F_2)$

As mass of the rod is M , hence acceleration of the rod is:

$$a = \frac{(F_1 - F_2)}{M} \qquad \text{....(i)}$$

If we now consider the motion of part AB of the rod [whose mass is equal to (M / L) y], then

$$F_1 - T = \frac{M}{L} y \times a$$

Where T is the tension in the rod at the point B.

Now, $$F_1 - T = \frac{M}{L} y \times \left(\frac{F_1 - F_2}{M} \right)$$

or $$T = F_1 \left(1 - \frac{y}{L} \right) + F_2 \left(\frac{y}{L} \right).$$

Alternative Method: Considering motion of the other part BC of the rod also, we can calculate tension at the point B. In this case,

$$T - F_2 = \frac{M}{L}(L - y) \times a$$

or $$T = F_2 + \frac{M}{L}(L - y) \times \frac{(F_1 - F_2)}{M}$$

$$= F_1 \left(1 - \frac{y}{L} \right) + F_2 \left(\frac{y}{L} \right).$$

17. D

Sol. $x = 3t - 4t^2 + t^3$

$$\frac{dx}{dt} = 3 - 8t + 3t^2 \text{ and } a = \frac{d^2x}{dt^2} = -8 + 6t$$

Now, $$W = \int F \, dx = \int ma \, dx = \int ma \frac{dx}{dt} dt$$

$$= \int_0^4 \frac{3}{1000} \times (-8 + 6t)(3 - 8t + 3t^2) dt$$

On integrating, we get; $W = 528 \, mJ$.

18. C

Sol. When force on the bullet is zero, then

$$0 = 600 - 2 \times 10^5 t$$

or $$t = \frac{600}{2 \times 10^5} = 3 \times 10^{-3} s$$

Now, $$I = \int_0^t F \, dt = \int_0^{3 \times 10^{-3}} (600 - 2 \times 10^5 t) dt$$

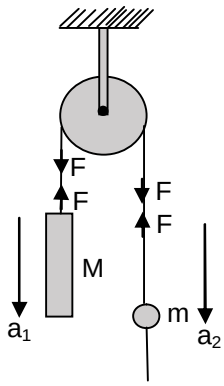
$$= \left[600t - \frac{2 \times 10^5 \times t^2}{2} \right]_0^{3 \times 10^{-3}}$$

$= \left[600 \times 3 \times 10^{-3} - 10^5 \times 9 \times 10^{-6}\right]$

$= \left[18 \times 10^{-1} - 0.9\right] = \left[1.8 - 0.9\right] = 0.9\ N - s.$

19. A

Sol. Let a_1 *and* a_2 be the accelerations of M *and* m respectively. Then



$Mg - F = Ma_1$ ....(i)

$mg - F = ma_2$ ....(ii)

Now, $1 + \frac{1}{2}a_2 t^2 = \frac{1}{2}a_1 t^2$

or $a_1 = \frac{2l}{t^2} + a_2$ ....(iii)

Solving equations (i), (ii) and (iii), we get;

$$F = \frac{2Mm}{(M-m)}\frac{l}{t^2}.$$

20. C

Sol. Let T be the tension in the string.

The upward force exerted on the clamp $= T\sin 30^o = \frac{T}{2}$

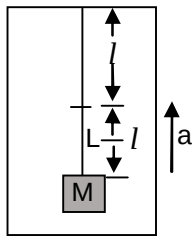
Given: $\frac{T}{2} = 40N$ *or* $T = 80N$ ....(i)

If a is the acceleration of monkey in upward direction,

$$a = \frac{T - mg}{m} = \frac{80 - 5 \times 10}{5} = 6m/s^2.$$

21. A

Sol. Let a be the acceleration of the lift.



Mass of lower portion of string

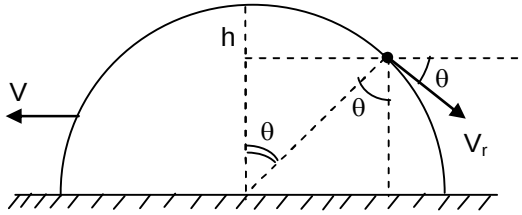
$= \frac{m}{L}(L - l)$

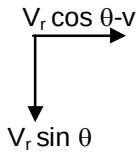
$\therefore \quad T - Mg - \frac{mg}{L}(L - l) = \left(M + \frac{m}{L}(L-1)\right)a$

$\therefore \quad a = \frac{T}{M + m - \frac{ml}{L}} - g$

22. A

Sol.





Let v_r be the velocity of particle relative to hemisphere and v be the linear velocity of hemisphere at this moment. Then from conservation of linear momentum, we have

$$4mv = m(v_r \cos\theta - v)$$

or $$5v = v_r \cos\theta$$

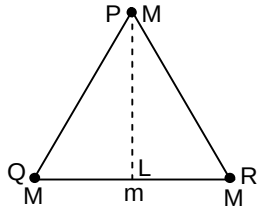
$$\therefore \quad v_r = \frac{5v}{\cos\theta}$$

$$\therefore \quad \omega = \frac{v_r}{R} = \frac{5v}{R\cos\theta}$$

23. B

Sol. Force on m due to masses at Q and R is zero. So, the net force is due to the mass at P.

Hence, $F = \dfrac{GMm}{(PL)^2}$



Now, $PL = l\sin 60^o = l\dfrac{\sqrt{3}}{2}$

$$\therefore \quad F = \frac{4GMm}{3l^2}.$$

24. A

Sol. $$g = \frac{GM}{r^2} = G.\frac{4}{3}\frac{\pi r^3 p}{r^2} = G.\frac{4}{3}\pi rp$$

$$\therefore \quad \frac{g_1}{g_2} = \frac{r_1 p_1}{r_2 p_2}.$$

25. C

Sol. Let r be the radius of each small droplets.

Then, volume of big drop $= 64 \times$ volume of each small droplet

$$\frac{4}{3}\pi R^3 = 64 \times \frac{4}{3}\pi r^3$$

$$R = 4r \qquad \text{....(i)}$$

Surface area of big drop $= 4\pi R^2$

Surface area of 64 small droplets $= 64 \times 4\pi r^2$

$\therefore$ Increase in surface area $= 64 \times 4\pi r^2 - 4\pi R^2$

$$= 4\pi\left[64r^2 - R^2\right]$$

$$= 4\pi\left[4R^2 - R^2\right]$$

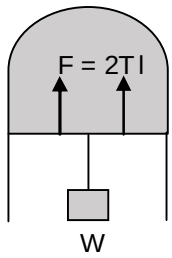
$$= 12\pi R^2$$

Energy needed $=$ surface tension $\times$ increase in surface area

$$= T \times 12\pi R^2 = 12\pi R^2 T.$$

26. D

Sol. The force of surface tension acting on the slider of length l in upward direction, balances the force due to the weight acting in downward direction.



$F = 2Tl = W$

$2T(0.3) = 1.5 \times 10^{-2}$

$\therefore\ T = \dfrac{1.5 \times 10^{-2}}{2 \times 0.3}$

$= 0.025\ N/m.$

27. C

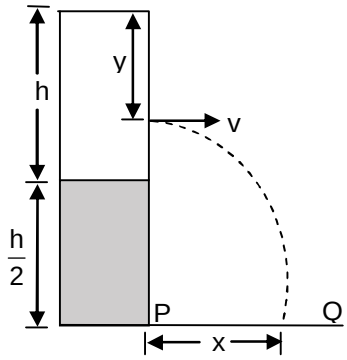
Sol. According to Stokes' law, the sphere encounters an opposing force F (or viscous force) when it falls through a viscous liquid of viscosity η with a velocity v and it is given by

$F = 6\pi\eta av$

Note : You can also see from the options, only **option (C)** has the dimensions of force.

28. B

Sol. Suppose there is a hole in the cylinder at depth y below water level.



Velocity of efflux $v = \sqrt{2gy}$

Time taken by water to reach on the plane PQ will be

$$t = \sqrt{\frac{2\left(\frac{h}{2} + (h-y)\right)}{g}} = \sqrt{\frac{3h-2y}{g}}$$

$\therefore$ Horizontal distance x travelled by the liquid is

$$x = vt = \sqrt{2gy}\sqrt{\frac{3h-2y}{g}}, \quad x = \sqrt{2y(3h-2y)}$$

For x to be maximum, $\dfrac{dx}{dy} = 0$

$$\frac{1}{2\sqrt{2y(3h-2y)}} \times 2(3h-4y) = 0$$

Or $3h - 4y = 0;\ \therefore\ y = \dfrac{3h}{4}$

Hence, x will be maximum at $y = \dfrac{3h}{4} = \left(h - \dfrac{h}{4}\right)$

Where, hole number 2 is present.

29. C

Sol. Energy radiated per second by the sun,

$E = \sigma T^4 4\pi R^2$

This energy falls uniformly on the inner surface of spheres centred around the sun. If r is the distance of the earth from the sun, then energy falling per second on unit area of the sphere of radius r is,

$\dfrac{2\pi R^2 \sigma T^4}{2\pi r^2} = \dfrac{\sigma R^2 T^4}{r^2}.$

It is only from the front hemisphere of the sun that the energy is falling on the earth and it is only on the front half of the earth.

Therefore, $\dfrac{4\pi R^2}{2}$ and $\dfrac{4\pi r^2}{2}$ have to be used.

The radiant power incident on the earth is given by:

$$Q = \pi r_0^2 \times \frac{\sigma R^2 T^4}{r^2} = \frac{\pi r_0^2 R^2 \sigma T^4}{r^2}.$$

30. C

Sol. Let length of open organ pipe $= l_0$

Length of closed organ pipe $= l_C$

Also, $l_0 = 2l_C$

Fundamental frequency of the open pipe,

$v_0 = 100$ Hz

Also, $v_0 = \frac{v}{2l_0} = 100$, $\frac{v}{l_0} = 200; \frac{v}{2\, l_C} = 200$, $\frac{v}{l_C} = 400$

Fundamental frequency of closed organ pipe

$$v_C = \frac{v}{4\, l_C} = \frac{400}{4} = 100 \text{ Hz}$$

So, the frequency of third harmonic of the closed organ pipe $= 3v_C = 300$ Hz.

Chemistry

PART – II

SECTION – A

31. D
Sol. Information.

32. D
Sol. See resonating structures.

33. B
Sol. Stereomers are not mirror images.

34. C
Sol. Information.

35. B
Sol. Symmetrical alkane with single free radical intermediate.

36. B
Sol. Kolbe's method.

37. B
Sol. Bromination followed by dehydrohalogenation.

38. D
Sol. $-CH_3$ is activating group
$-NO_2$ is stongly deactivating group
–Cl is weekly deactivating group

39. D

40. A
Sol. (a) (I) 0.5 mole O_3 = 24 g O_3
(II) 0.5 g atom of oxygen = 8 g
(III) $\frac{3.011\times10^{23}}{6.022\times10^{23}}\times32=16$ g O_2
(IV) $\frac{5.6}{22.4}\times44$ g $CO_2=11$ g CO_2

41. B
Sol. 10 g of $D_2O \equiv (2+8)N_A$ neutrons.

42. D
Sol. $v \propto \frac{Z}{n}$

43. D
Sol. Orbital angular momentum $=\sqrt{l(l+1)}\frac{h}{2\pi}$ $l=1$ for p - orbital.

44. D

Sol. $PV = n_{air}RT$

$$n_{oxygen} = \frac{14}{100}n_{air}$$

45. C

Sol. Because the number of moles is constant.

$$\frac{P_iV_i}{T_i} = \frac{P_fV_f}{T_f}; P_f = \frac{P_iV_iT_f}{V_fT_i}$$

$$P_f = \frac{P_iV_iT_f}{V_fT_i}$$

$$= 3.21\times10^5 \text{ Pa} \times \frac{V_i}{1.03V_i} \times \frac{(273+28.0)}{(273-5.00)}$$

$$= 3.50\times10^5 \text{ Pa}$$

46. D

Sol. $\Delta U = q + w,\ w > 0$ as work is done on the system.

47. B

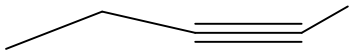
Sol. $|\text{Work done}|$ is area under the P-V curve.

48. D

Sol. $K_{eq} = \frac{k_f}{k_b} = \frac{\text{Conc. of proudcts as per rate law}}{\text{Conc. of reactants as per rate law}}$, stoichiometry of reaction is required.

49. D

Sol. $Mg_2C_3 \xrightarrow{4H_2O} 2Mg(OH)_2 + C_3H_4$

$C_3H_4 \xrightarrow[\text{ii) } C_2H_5Br]{\text{i) Na}}$ 

50. A

Sol. Weakest acid has strongest conjugate base and vice versa.

51. D

Sol. $pH + pOH = pK_w$

52. B

Sol. Information.

53. B

Sol. $\text{Size} \propto \frac{1}{z_{eff.}}$

54. D

Sol. SF_4: sp^3d hybridization
CF_4: sp^3 hybridization
XeF_4: sp^3d^2 hybridization

55. B

56. D
Sol. Atomic radius increases due to addition of extra shell.

57. B
Sol. $Na + (x+y)NH_3 \longrightarrow Na(NH_3)_x + \underset{(blue)}{e^-(NH_3)_y}$

58. C
Sol. Information (Borax bead: $NaBO_2 + B_2O_3$)

59. C
Sol. Inert pair effect makes +2 as more stable than +4 oxidation state.

60. A
Sol. If $Q > K$, equilibrium shifts backwards.

Mathematics

PART – III

SECTION – A

61. C

Sol. The $(r+1)$th term in the expansion of $[(3/2)x^2 - (1/3x)]^9$ is given by

$$T_{r+1} = {}^9C_r \left(\frac{3}{2}x^2\right)^{9-r}\left(-\frac{1}{3x}\right)^r = {}^9C_r(-1)^r \frac{3^{9-2r}}{2^{9-r}}x^{18-3r} \quad \ldots(1)$$

Since we are looking for the coefficient of the term independent of x in the expansion of

$$(1 + x + 2x^3)\left(\frac{3}{2}x^2 - \frac{1}{3x}\right)^9 \quad \ldots(2)$$

we must get the coefficient of x^0, x^{-1} and x^{-3} in the expansion of $[(3/2)x^2 - (1/3x)]^9$. For x^0, r must be 6 in (1); for x^{-1}, there is no value of r ; and for x^{-3}, r must be 7 in (1). Therefore, the coefficient of the term independent of x in (2) is

$$1.{}^9C_6(-1)^6.\frac{3^{9-12}}{2^{9-6}} + 2.{}^9C_7(-1)^7.\frac{3^{9-14}}{2^{9-7}}$$

$$= \frac{9.8.7}{1.2.3}.\frac{3^{-3}}{2^3} + 2.\frac{9.8}{1.2}(-1).\frac{3^{-5}}{2^2} = \frac{7}{18} - \frac{2}{27} = \frac{17}{54}$$

62. D

Sol. We have

$$(1-x)^n = C_0 - C_1x + C_2x^2 - C_3x^3 + \ldots$$

$$\Rightarrow \quad x(1-x)^n = C_0x - C_1x^2 + C_2x^3 - C_3x^4 + \ldots$$

$$\int_0^1 x(1-x)^n\,dx = \int_0^1 x(1-x)^n\,dx = \int_1^0 (1-t)t^n(-1)dt \qquad [\text{Put } 1-x=t]$$

$$= \int_0^1 \left(t^n - t^{n+1}\right)dt = \left|\frac{t^{n+1}}{n+1} - \frac{t^{n+2}}{n+2}\right|_0^1$$

$$= \frac{1}{n+1} - \frac{1}{n+2} = \frac{1}{(n+1)(n+2)}$$

Integrating R.H.S of (1) we get $\left(\frac{C_0x^2}{2} - \frac{C_1x^3}{3} + \frac{C_2x^4}{4} - \ldots\right)\Big|_0^1$

$$= \frac{C_0}{2} - \frac{C_1}{3} + \frac{C_2}{4} - \ldots$$

Thus $\frac{C_0}{2} - \frac{C_1}{3} + \frac{C_2}{4} - \ldots = \frac{1}{(n+1)(n+2)}$

63. C

Sol. Here, radius $\sqrt{\left(\frac{\lambda}{2}\right)^2 + \left(\frac{1-\lambda}{2}\right)^2 - 5} \le 5 \quad \Rightarrow 2\lambda^2 - 2\lambda - 119 \le 0$

$$\therefore \frac{1-\sqrt{239}}{2} \le \lambda \le \frac{1+\sqrt{239}}{2} \Rightarrow -7.2 \le \lambda \le 8.2 \text{ (nearly) } \therefore \lambda = -7, -6, \ldots, 8$$

64. C

Sol. Let mid point be (h, k), then equation of chord is $hx + ky = h^2 + k^2$

Now on homoginising $x^2 + y^2 = 4$ with the help of this chord

$$x^2 + y^2 = 4\left(\frac{hx+ky}{h^2+k^2}\right)^2$$

Now as subtended angle is 90° $\Rightarrow$ Coeff. of x^2 + Coeff. of $y^2 = 0$

$1 - \frac{4h^2}{\left(h^2+k^2\right)^2} + 1 - \frac{4k^2}{\left(h^2+k^2\right)^2} = 0$, represents a pair of straight lines passing through origin and intersection points of circle and chords.

$2(x^2 + y^2)^2 = 4(x^2 + y^2) \Rightarrow x^2 + y^2 = 2$

65. D

Sol. The equation of a chord joining points having eccentric angles α and β is given by

$$\frac{x}{a}\cos\left(\frac{\alpha+\beta}{2}\right) + \frac{y}{b}\sin\left(\frac{\alpha+\beta}{2}\right) = \cos\left(\frac{\alpha-\beta}{2}\right)$$

If it passes through $(ae, 0)$ then

$$e\cos\left(\frac{\alpha+\beta}{2}\right) = \cos\left(\frac{\alpha-\beta}{2}\right)$$

$$\Rightarrow \quad e = \frac{\cos\left(\frac{\alpha-\beta}{2}\right)}{\cos\left(\frac{\alpha+\beta}{2}\right)} \Rightarrow \frac{2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)}{2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha+\beta}{2}\right)} \Rightarrow e = \frac{\sin\alpha+\sin\beta}{\sin(\alpha+\beta)}$$

66. C

Sol. Since $3(3)^2 + 5(5)^2 - 32 > 0$. So the given point lies outside the ellipse. Hence, two real tangents can be drawn from the point to the ellipse.

67. B

Sol. We know that the product of perpendiculars from two foci of an ellipse upon any tangent is equal to the square of the semi-minor axis.

$(S_1 F_1)(S_2 F_2) = 3$.

68. D

Sol. Normal at t_1 meets the curve again at t_2.

So normal passes through point $(ct_1, c/t_1)$ and $(ct_2, c/t_2)$.

Slope of normal $= \frac{c/t_2 - c/t_1}{ct_2 - ct_1} = -\frac{1}{t_1 t_2}$

Slope of normal from equation of normal at $t_1 = t_1^2$

So $t_1^2 = -\frac{1}{t_1 t_2} \Rightarrow t_1^3 t_2 = -1$

69. D

Sol. Let the coordinates of P be (α, β)

Then, PQ = 2β and OP = $\sqrt{\alpha^2+\beta^2}$

Since OPQ is an equilateral triangle OP = PQ

$\Rightarrow \quad \alpha^2 + \beta^2 = 4\beta^2 \Rightarrow \alpha^2 = 3\beta^2 \quad \Rightarrow \quad \alpha = \pm\sqrt{3}\beta$

Also since (α, β) lies on the given hyperbola, $\frac{\alpha^2}{a^2} - \frac{\beta^2}{b^2} = 1$

$\Rightarrow \quad \frac{3\beta^2}{a^2} - \frac{\beta^2}{b^2} = 1 \Rightarrow \frac{3}{a^2} - \frac{1}{b^2} = \frac{1}{\beta^2} > 0 \quad \Rightarrow \frac{b^2}{a^2} > \frac{1}{3} \quad \Rightarrow e^2 - 1 > \frac{1}{3}$

$\Rightarrow \quad e^2 > \frac{4}{3} \quad \Rightarrow e > \frac{2}{\sqrt{3}}$

70. A

Sol. Equation of focal chord:

$y = m(x - 4)$

Thus $\frac{|0 - m(6-4)|}{\sqrt{1+m^2}} = \sqrt{2} \quad \Rightarrow \quad \frac{4m^2}{1+m^2} = 2 \quad \Rightarrow m = \pm 1$

71. B

Sol. $y = mx - 2am - am^3$ (1)

$y = m(x - b) - 2cm - cm^3 \Rightarrow y = mx - (b + 2c)m - cm^3$ (2)

from (1) – (2) we gets, $0 = (b + 2c - 2a)m + (c - a)m^3$

$\Rightarrow \quad m\{(b + 2c - 2a) + (6 - a)m^2\} = 0$

Since common normal other than x–axis i.e., $m \neq 0$

$\therefore \quad m^2 = \frac{2(a-c)-b}{(c-a)}$

$= -2 + \frac{b}{a-c} > 0 \Rightarrow \frac{b}{a-c} > 2$

72. B

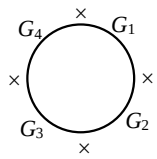
Sol. First make the 4 girls sit around a circular table.

This can be done in $(4 - 1)! = 3! = 6$.

After that '3' boys can occupy any three crosses.

This can be done in ${}^4P_3 = 4!$ ways.

Hence total no. of ways $= 3! \times 4!$.



73. B

Sol. $xyz = 3 \times 2^3 \times 5^3$

Now the power of '3' is 1, then 3 can be distributed among three variables in '3' ways. Simmilarly the power of '2' is 2, hence 2 can be distribute among x, y and z in '10' ways and same is the case of 5^3.

Hence total ways $= 3 \times 10 \times 10 = 300$.

74. B

Sol. Any divisor of $2^5.3^4.5^2$ is of the form $2^a.3^b.5^c$ where $0 \le a \le 5$, $0 \le b \le 4$ and $0 \le c \le 2$.

Hence, the sum of the divisors

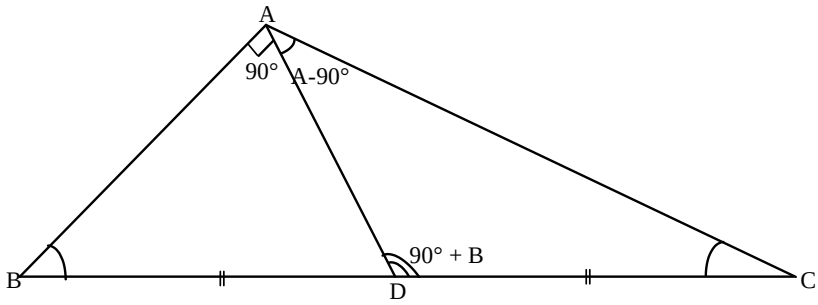
$= \sum_{a=0}^{5}\sum_{b=0}^{4}\sum_{c=0}^{2} 2^a 3^b 5^c = (1 + 2 + \ldots + 2^5)(1 + 3 + \ldots + 3^4)(1 + 5 + 5^2)$

$= \left(\frac{2^6 - 1}{2 - 1}\right)\left(\frac{3^5 - 1}{3 - 1}\right)\left(\frac{5^3 - 1}{5 - 1}\right)$

$= 63 \times 121 \times 31 = 3^2.7^1.11^2.31$

75. B

Sol. Since AD is the median



$\therefore$ BD : DC = 1 : 1. From Figure, $\angle$ADC = 90° + B.

Now, applying m : n rule in triangle ABC, we get

(1 + 1) cot(90° + B) = 1.cot90° – 1.cot(A –90°)

or, –2tan B = 0 – cot[–(90° – A)]

or, –2tan B = –[–cot(90° – A)]

or, –2tanB = tanA $\quad\therefore$ tanA + 2tanB = 0.

76. A

Sol. We have, cosA cosB + sinA sinB sinC = 1

from the given relation

$\sin C = \dfrac{1-\cos A\cos B}{\sin A\sin B} \le 1$ …(i)

$\Rightarrow 1 \le \cos A\cos B + \sin A\sin B$

$\Rightarrow 1 \le \cos(A - B)$ (as $\cos(A - B) \not> 1$)

$\Rightarrow 1 = \cos(A - B) \Rightarrow A - B = 0 \Rightarrow A = B$ …(ii)

$\therefore$ from (i)

$\sin C = \dfrac{1-\cos^2 A}{\sin^2 A} = 1$

$\therefore$ C = 90° and A + B = 90°

or A = B = 45° (by (ii))

By sine Law,

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$\frac{a}{\frac{1}{\sqrt{2}}} = \frac{b}{\frac{1}{\sqrt{2}}} = \frac{c}{1}$$

$a : b : c = 1 : 1 : \sqrt{2}$

77. A

Sol. $\cos\dfrac{\pi}{7}\cos\dfrac{2\pi}{7}\cos\dfrac{3\pi}{7} = \cos\dfrac{\pi}{7}\cos\dfrac{2\pi}{7}\cos\left(\pi - \dfrac{4\pi}{7}\right) = -\cos\dfrac{\pi}{7}\cos\dfrac{2\pi}{7}\cos\dfrac{4\pi}{7}$

$$= -\left[\frac{\sin\left(2^3.\frac{\pi}{7}\right)}{2^3\sin\frac{\pi}{7}}\right] = -\frac{\sin\frac{8\pi}{7}}{8\sin\frac{\pi}{7}} = \frac{1}{8}$$

78. A

Sol. Let $S = \cos 2\theta + \cos\theta$

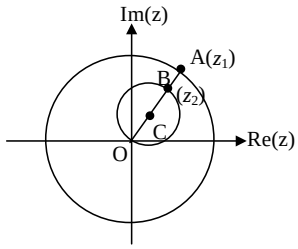
$= 2\cos^2\theta - 1 + \cos\theta = -1 + 2\left(\cos^2\theta + \frac{1}{2}\cos\theta + \frac{1}{16}\right) - \frac{1}{8}$

$= -\frac{9}{8} + 2\left(\cos\theta + \frac{1}{4}\right)^2 \geq -\frac{9}{8}$

So, the minimum value $S = -9/8$

79. B

Sol. Shortest distance = OA – OB = 12 – 10 = 2



80. D

Sol. Put $x=r\cos\theta$ and $y=r\sin\theta$ then find product of roots in cubic equation in r.

81. A

82. A

Sol. $(b+c)^{-1}$, $(c+a)^{-1}$, $(a+b)^{-1}$ are in A.P so by using properties of AP, we can show that $2b^2 = a^2+b^2$

83. B

Sol. Total number of +ve integral solution =30 and there are 4 cases to be +ve
i.e (+,+,+),(-,-,+)(+,-,-),(-,+,-)
Hence total no. of +ve integral solution= 120.

84. A

Sol. The $(r + 1)$th term in the expansion of $\left(\sqrt{\frac{x}{3}} + \frac{3}{2x^2}\right)^{10}$ is given by

$T_{r+1} = {}^{10}C_r \left(\sqrt{\frac{x}{3}}\right)^{10-r} \left(\frac{3}{2x^2}\right)^r = {}^{10}C_r \frac{x^{5-(r/2)}}{3^{5-(r/2)}} \cdot \frac{3^r}{2^r x^{2r}}$

$= {}^{10}C_r \frac{3^{(3r/2)-5}}{2^r} x^{5-(5r/2)}$

For T_{r+1} to be independent of x, we must have $5 - (5r/2) = 0$ or $r = 2$. Thus, the 3rd term is independent of x and its coefficient is given by

${}^{10}C_2 \frac{3^{3-5}}{2^2} = \frac{10\times 9}{2} \times \frac{3^{-2}}{4} = \frac{5}{4}.$

85. A

Sol. As a, b, c are in H.P. 1/a, 1/b, 1/c are in A.P.

$\Rightarrow \quad \frac{1}{b} - \frac{1}{a} = \frac{1}{c} - \frac{1}{b} \; or \; \frac{2}{b} = \frac{1}{a} + \frac{1}{c} \qquad \ldots.(1)$

Therefore,

$$\left(\frac{1}{b}+\frac{1}{c}-\frac{1}{a}\right)\left(\frac{1}{c}+\frac{1}{a}-\frac{1}{b}\right)=\left(\frac{1}{b}+\frac{1}{c}+\frac{1}{c}-\frac{2}{b}\right)\left(\frac{1}{c}+\frac{1}{b}-\frac{1}{c}\right)$$ [using (1)]

$$=\left(\frac{2}{c}-\frac{1}{b}\right)\left(\frac{1}{b}\right)=\frac{2}{bc}-\frac{1}{b^2}.$$

86. A

Sol. We have

$t_r = 1^2 + 3^2 + 5^2 + \ldots + (2r-1)^2$

$$= \sum_{k=1}^{r}(2k-1)^2 = \sum_{k=1}^{r}\left(4k^2-4k+1\right)$$

$$= 4\left\{\frac{1}{6}r(r+1)(2r+1)\right\}-\frac{4}{2}r(r+1)+r$$

$$= \frac{1}{6}r\left[4(r+1)(2r+1)-12(r+1)+6\right]$$

$$= \frac{1}{6}r\left[8r^2-2\right]=\frac{1}{3}\left(4r^3-r\right)$$

Thus, sum of n terms of the given series

$$\frac{4}{3}\sum_{r=1}^{n}r^3-\frac{1}{3}\sum_{r=1}^{n}r=\frac{4}{3}.\frac{1}{4}n^2(n+1)^2-\frac{1}{3}.\frac{1}{2}n(n+1)$$

$$= \frac{1}{6}n(n+1)\left[2n(n+1)-1\right]=\frac{1}{6}n(n+1)\left(2n^2+2n-1\right)$$

87. A

Sol. A.M. $\geq$ G.M.

$$\Rightarrow \quad \frac{a_1+a_2+\ldots+a_{n-1}+2a_n}{n} \geq (a_1.a_2 \ldots a_{n-1}.2a_n)^{1/n}$$

$$\Rightarrow \quad a_1 + a_2 + \ldots + 2a_n \geq n(2c)^{1/n}$$

88. B

Sol. Here the equation is $x^2 - (10 + a)x + 10a + 1 = 0$ (1)

Since integral roots will always be rational it means D should be a perfect square.

From (1) $D = a^2 - 20a + 96$

$\Rightarrow$ $D = (a - 10)^2 - 4$

$\Rightarrow$ $4 = (a - 10)^2 - D$

If D is a perfect square it means we want difference of two perfect square as 4 which is possible only when $(a - 10)^2 = 4$ and $D = 0$

$\Rightarrow (a - 10) = \pm 2$

$\Rightarrow a = 12, 8$

89. A

Sol. Sum & product of the roots,

$$\sin\alpha + \cos\alpha = -\frac{b}{a}, \quad \sin\alpha\cos\alpha = \frac{c}{a}$$

$$\because \quad 1 = \sin^2\alpha + \cos^2\alpha = (\sin\alpha + \cos\alpha)^2 - 2\sin\alpha\cos\alpha = \frac{b^2}{a^2} - \frac{2c}{a}$$

$\Rightarrow \quad a^2 = b^2 - 2ac \Rightarrow a^2 - b^2 + 2ac = 0$
$\Rightarrow a^2 + c^2 + 2ac = b^2 + c^2 \Rightarrow (a + c)^2 = b^2 + c^2$

90. C

Sol. $\alpha + \beta = \frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{\alpha^2 + \beta^2}{\alpha^2\beta^2} = \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha^2\beta^2}$

$\Rightarrow -\frac{b}{a} = \frac{b^2/a^2 - 2c/a}{c^2/a^2} = \frac{b^2 - 2ac}{c^2} \Rightarrow \frac{2a}{c} = \frac{b^2}{c^2} + \frac{b}{a} = \frac{ab^2 + bc^2}{ac^2}$

$\Rightarrow 2a^2c = ab^2 + bc^2 \Rightarrow \frac{2a}{b} = \frac{b}{c} + \frac{c}{a}$ [dividing by abc] $\Rightarrow \frac{c}{a}, \frac{a}{b}, \frac{b}{c}$ are in A.P.

$\Rightarrow \frac{a}{c}, \frac{b}{a}, \frac{c}{b}$ are in H.P.